

Aiden Song

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Educational Experience

Shanghai High School International Division (G7-12)	Scholarship + Top 7%	2022.9 – 2028.6
Carnegie Mellon Uni:	Concepts of Math, (96/100) and Imperative Computing (88/100)	2025.6 – 2025.8
Stanford Uni:	Introduction to Data Science	2024.7 – 2024.8
Standardized Test:	SAT 1550 (800M/750EBWR), TOEFL 119 (30L/30R/29S/30W)	

Awards & Honors

- **S.T.Yau Science Award**, Global Top 10 & **Sichuan Science Fair**, Top 9 2025-2026
Topic: Efficient Density Driven Flow Simulation
- **Physics Bowl D1**, Gold(2025) **D2**, Silver(2026) 2025-2026
- **John Locke Essay Competition in Public Policy**, Shortlisted 2026.7
Topic: Discounts Rate for Environmental Policies in the Long-run
- **USA Computing Olympiad (USACO)**, Gold 2026.2
- **International Math Modeling Challenge (IMMC)**, Intl Finalist, 2x Regional Finalist 2025.12
Topics: Wild Life Protection, Cost of Solar Energy Production, and Predicting Luck
- **High School Mathematical Contest in Modeling (HiMCM)**, Finalist (Top 21 / 585) 2025.11
Topic: Smoke&Fire-aware Coordinated Optimal Uncertainty-informed Traversal
- **British Physics Olympiad**, Gold 2025.11
- **American Invitational Mathematics Examination (AIME)**, Scored 9 Points 2024.2

Oceanography Experiences

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| Yangtze River Expedition | <i>Field Researcher</i> | <i>112hr/wk, 2wk/yr</i> | 2026.6–2026.7 |
| <ul style="list-style-type: none">• Collected and analyzed data from water samples for Yangtze River from high-altitude headwaters toward downstream regions and interviewed 50+ local residents, workers, and stakeholders to investigate how hydropower development, climate change, tourism, etc. reshape communities along the river basin• Website in creation | | | |
| US-China Colab on Climate Solutions | <i>Student</i> | <i>5hr/wk, 10wk/yr</i> | 2026.3–2026.5 |
| <ul style="list-style-type: none">• Collaborated with American/Chinese students on climate policy and sustainability challenges through research, dialogue, and joint project development at Stanford (only 15 ppl were chose from China) and completed an online media campaign w/ website on Coastal Protection Day | | | |
| Research: Efficient Density Driven Flow Simulation | | <i>30hr/wk, 32wk/yr</i> | 2024.4–2026.2 |
| <ul style="list-style-type: none">• Developed a 0-to-1 method for reconstructing density-driven flows from sparse data to improve prediction of ocean currents and other geophysical flows where dense observational data are costly or unavailable• Combined physics-informed neural networks with governing physical equations to cut the data required for a plausible simulation to ~1%, versus the ~30% needed by prior methods• Selected in China's National Top Talent Program for Computer Science and conducted research with Prof. Yang Bin at East China Normal University to enhance physics-informed neural networks (PINNs), building broader foundation in deep learning and AI; recognized as Outstanding Student (Top 33/1,300+) nationally | | | |
| Research: Slowing of the Global Ocean Conveyor Belt | | <i>10hr/wk, 32wk/yr</i> | 2023.7–2024.12 |
| <ul style="list-style-type: none">• Conducted data analysis on oceanographic records with an original physical model to investigate the weakening of the Atlantic Meridional Overturning Circulation and identified freshwater input from melting glaciers as the core driver disrupting density gradient and the overall climate system | | | |
| Research: Zebrafish Caudal Fin Regeneration | | <i>10hr/wk, 20wk/yr</i> | 2023.7–2024.3 |

- Selected by Chinese Academy of Sciences to design and track caudal-fin regeneration in different temperatures and identified 28°C as the optimal for repair and authored a 7000-word report that won 2nd place in the Shanghai Science Association for Young Talent
- Maintained zebrafish housing, performed cryosectioning to prepare fin-tissue samples, and operated specialized devices like stereomicroscope and cryostat to image and analyze fin regrowth

Social Awareness Contributions

Alphadeer AI Education *Founder/Developer* *5hr/wk, 32wk/yr* 2025.3–Now

- Led 30+ member to promote WeChat public account, and website development by networking UNESCO China staff and other professionals to secure speakers for 10+ online seminars on trends in climate change
- Translated the UNESCO AI Education Student Framework on AI education (80-pg report) into Chinese to increase accessibility for Chinese students and educators
- Authored 5k+ word literature review on modern AI education's global impact on learning with the University of Macau by conducting pedagogical research and synthesizing 80+ academic sources
- Exhibition (open in Sept)

Math Modeling Club *President* *4hr/wk, 32wk/yr* 2023.9–Now

- Led 100+ member by designing and running weekly training sessions on mathematical modeling techniques, writing starter code and templates to guide them on model selection and strategy
- Invited expert speakers and provided 1-on-1 consulting for HiMCM, IMMC, and the HKU × SHSID competition, helping members earn outstanding, finalist, meritorious placement in contests

SHSID Glacier Week *Project Manager* *20hr/wk, 5wk/yr* 2026.3–2026.5

- Led 8 members to organized a four-day K-12 climate-education exhibition across 3 campus sites for 84+ visiting classes and won Global 1st in Sea Beyond, securing €5,000 sponsorship
- Designed multi-format exhibits to appeal to different visitors: informational infographics, hands-on art piece installation simulating a melting-glacier with polar-animal models, a food-based display linking sea-level rise to local Shanghai cuisine, and a coded online trivia game (1,000+ lines)
- Launched the "Frozen Voices" social media account across social medial and produced research-backed educational videos on topics from albedo feedback to AMOC that reached 10k+ total viewers

Service for Ocean *High School Rep* *20hr/wk, 1wk/yr* 2023.7–2024.7

- Made public speech on marine conservation education to 100+ peers and educators to advocated ocean protection initiatives on the Ocean Decade project by UN, raising eco-awareness among youth
- Docent for informational section at the local Ocean park.

Personal Interests

Research Internship *Intern* *40 hr/wk, 8 wk/yr* 2026.6–2026.8

- Took ownership of a large ongoing computational oceanography and machine-learning project, independently reconstructing its data, codebase, and experimental history.
- Prof. Jianzhong Ge's Research Group, East China Normal University
- Developed and evaluated deep-learning forecasting systems based on MeshGraphNets, designed controlled experiments, and managed large-scale GPU training across multiple computing servers.
- Analyzed model performance under ordinary and extreme conditions and translated findings into scientific figures, presentations, and an IEEE-style research manuscript.

Online Rock Museum *Founder* *8hr/wk, 32wk/yr* 2025.12–Now

- Founded and built Shishijie, an interactive 3D museum preserving a family collection of 200+ naturally patterned Yangtze stones; designed rotatable exhibits, dynamic lighting, geological and cultural narratives, and a public stone-upload archive.

Crescent Philharmonic Orchestra *Founder/Leader* *5hr/wk, 32wk/yr* 2024.9–Now

- Ended a long-standing divide between the string and wind instrument programs at school and formed the school's first full orchestra with an ensemble of 100+ active members after much negotiation
- Lead 30+ volunteer musicians in weekly community performances at Ruijin Hospital and Youyouxin Nursing Home, contributing 100+ volunteer hours of live music for patients and elderly residents
- Organized and led orchestra concert for 500+ audience and performed at the school art festival and the school's first student-led concert, integrating orchestra into school culture and tradition

Kaggle *Competitor* *4hr/wk, 52wk/yr* 2024.7–Now

- Build end-to-end modeling pipelines (feature engineering, model ensembling, cross-validation, post-processing) and shared reproducible notebooks, solution write-ups, and reusable code for others to build on
- Placed 260/62092 in open-source coding with 500+ upvotes, 400+ forks, 1 gold, 3 silver & 12 bronze medals
- Received Scientific Forgery Detection 61/1564 Top 4% Silver medal, BirdClef 2026 Silver Medal, Neurogolf 2026 Silver Medal, ROGII Geology prediction Bronze Medal, Diabetes prediction challenge Top 4%, Road accident risk Top 13%, Deep Past Challenge Top 16%

Badminton Varsity Team *Member* *3hr/wk, 32wk/yr* 2023.9–Now

- Trained and competed in interschool matches